Skills Summary

Statistical Questions, Centre and Spread

Use this reference while completing your worksheet or when studying.

Skill 1 — Is it a statistical question?

A question is statistical when you expect the answers to vary. It is about the *answers*, not the topic.

Concrete test: collect the answers and find the range. Range = 0 means every answer was the same, so nothing varied and the question was not statistical. A question about *one* person or thing almost never varies; a question about a *group* usually does.

Example: Is "How many minutes did the students in my group read last night?" statistical? Answers collected: $10, 25, 25, 40$.
Step 1. Whether a question is statistical is decided by variability in its answers, so the thing to compute is the range of what was collected.
Step 2. Largest 40 minus smallest 10.
$\Rightarrow$ range = **30**, not zero, so **yes**, it is statistical

Try it: Is "How many minutes did Mr Diaz read last night?" statistical? Answers collected from four people: $25, 25, 25, 25$.

check: range = **0**, so no — one person, one answer

Skill 2 — Centre: mean and median

Mean = (add all the values) $\div$ (how many values). **Median** = the middle value once the data is *sorted*; with an even number of values, halfway between the middle two.

Sort first, always. And when one value is far from the rest, the mean chases it while the median barely moves — that is why both are worth quoting.

Example: Find the mean and the median of $4, 5, 7, 8, 16$.
Step 1. Both are centres, but they answer different questions, so compute each with its own rule rather than expecting them to agree.
Step 2. Mean: $(4 + 5 + 7 + 8 + 16) \div 5 = 40 \div 5$. Median: already sorted, five values, so the middle one is the third.
$\Rightarrow$ mean = **8**, median = **7**

Try it: Find the mean and the median of $3, 6, 9, 10, 22$.

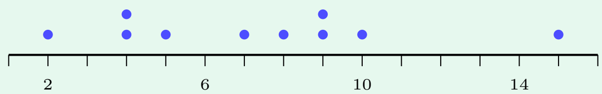
check: mean = **10**, median = **9**

Skill 3 — Spread: range and IQR

Range = largest − smallest. It uses only the two extremes, so one unusual value controls it.

Interquartile range describes the middle half instead. Sort the data, split it into a lower half and an upper half, take the median of each: those are Q_1 and Q_3, and $\text{IQR} = Q_3 - Q_1$. With an odd number of values, leave the overall median out of both halves.

Example: Range and IQR of $2, 4, 4, 5, 7, 8, 9, 9, 10, 15$.



Step 1. Two spreads are wanted and they measure different things — the range reads the two ends of the plot, the IQR reads only the middle half — so take them separately.
Step 2. Range: $15 - 2$. Halves of five: $2, 4, 4, 5, 7$ gives $Q_1 = 4$, and $8, 9, 9, 10, 15$ gives $Q_3 = 9$, so $\text{IQR} = 9 - 4$.
$\Rightarrow$ range = **13**, IQR = **5**

Try it: Range and IQR of $5, 6, 6, 7, 9, 10, 12, 13, 14, 20$.

check: range = **15**, IQR = **7**

Skill 4 — Chance from a data set

Probability = (how many values fit the description) ÷ (how many values there are altogether), written as a fraction in lowest terms.

Read the words exactly: "more than 9" excludes a value of 9, "at least 9" includes it. Count the dots on the plot rather than the labels on the axis.

Example: One of the ten values above is chosen at random. Find $P(\text{value is at least } 9)$.
Step 1. Every value is equally likely to be chosen, so the probability is just a count of the values that fit, over the ten values there are.
Step 2. At least 9 means $9, 9, 10, 15$ — four of the ten — and $4/10$ reduces.
$\Rightarrow$ $P = \mathbf{\frac{2}{5}}$

Try it: From the same ten values, find $P(\text{value is at most } 4)$.

check: $\mathbf{\frac{3}{10}}$